Enhancing Unmanned Aerial Vehicles Path Planning: A Multi-Heuristic A* Approach with Adaptive Optimization for Dynamic Environments

Min-Fan Ricky Lee, Ferreira Roig Ocampos Saul & Vazquez Medina Luis Eduardo

***Abstract*—Unmanned Aerial Vehicles (UAVs) require intelligent and adaptive path planning to navigate complex and dynamic environments while ensuring flight safety. Traditional A* algorithms, though effective in static settings, struggle with high computational overhead and limited responsiveness to unforeseen obstacles. While recent enhancements based on predictive path planning offer partial improvements, they often remain insufficient for real-time applications. To address these limitations, we propose an enhanced Multi-Heuristic A* (MH-A*) algorithm that dynamically integrates multiple heuristic functions to optimize both global and local trajectory planning. The approach incorporates Bresenham's algorithm for real-time obstacle detection and efficient key node extraction, resulting in smoother and more responsive navigation. Additionally, a dual-layer optimization framework enables global planning to be refined through adaptive local adjustments. Results indicate that the proposed method significantly reduces path length, improves real-time obstacle avoidance, and lowers computational demand compared to existing solutions. These advancements position the MH-A* algorithm as a promising candidate for autonomous UAV navigation in mission-critical scenarios such as urban air mobility, search-and-rescue operations, and disaster response. Future work includes empirical validation through benchmarking against state-of-the-art algorithms.**

***Index Terms*—Heuristic algorithms, Trajectory optimization, Unmanned aerial vehicles.**

I. Introduction

UNMANNED Aerial Vehicles (UAV) encounter significant challenges when operating in dynamic and unstructured environments. Traditional algorithms like A* and DWA (Dynamic Window Approach), though widely used, often suffer from high computational overhead, path redundancy, and poor adaptability to dynamic changes—limitations that jeopardize mission efficiency and UAV safety.

Efficient path planning is critical in UAV applications such as disaster response, surveillance, search and rescue, and urban air mobility. These missions demand not only optimal routes but also rapid adaptability to unexpected environmental changes. Ensuring safety, minimizing energy consumption, and supporting real-time decision-making are essential to mission success.

Although recent work, such as the predictive path planning approach by He et al. [1] [2] [3], introduces valuable enhancements, existing methods continue to exhibit high computational costs and insufficient responsiveness to dynamic environments. These limitations highlight a gap in the development of scalable, lightweight algorithms capable of balancing global optimization with real-time adaptability.

This work builds upon and extends prior strategies, particularly those focused on predictive path planning like A* algorithm [1] and fusion algorithm [4]. Unlike methods that rely on precomputed obstacle trajectories or static and dynamic modeling, the proposed approach emphasizes adaptability and low computational burden without sacrificing trajectory quality and improves metrics in path planning.

To address these gaps, we propose an enhanced Multi-Heuristic A* (MH-A*) algorithm that introduces:

- **Dynamic heuristic selection** based on environmental complexity and UAV operational constraints.
- **Real-time obstacle detection** using Bresenham's algorithm to refine paths on-the-fly.
- **A dual-layer optimization framework** to combine global planning with adaptive local adjustments.

The proposed MH-A* approach introduces the following key innovations:

1) **Multi-Heuristic Search Optimization:** Unlike conventional A*, which uses a fixed heuristic, MH-A* dynamically selects the most appropriate heuristic function [5], depending on environmental

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conditions, reducing unnecessary node expansion and improving efficiency.

2) **Obstacle-Aware Node Extraction:** Bresenham's algorithm [6] is employed to detect and react to obstacles in real time, enabling precise path correction and reduced computational redundancy.
3) **Dual-Layer Optimization:** The planning framework comprises:
   - **Global Layer:** Generates an initial optimal path using MH-A*.
   - **Local Layer:** Refines the path in real time using an adaptive evaluation function that reacts to dynamic environmental changes.

By integrating predictive efficiency with real-time adaptability, the MH-A* framework aims to outperform current state-of-the-art approaches, achieving shorter, smoother, and more computationally efficient trajectories for UAVs in dynamic environments.

II. Method

A. Overview of the Proposed Algorithm

The proposed path planning framework enhances the conventional A* algorithm and fusion algorithm through dynamic heuristic switching, obstacle-aware node refinement, and a dual-layer optimization structure. The methodology focuses on improving both computational efficiency and real-time adaptability for UAVs navigating in complex, dynamic environments.

B. Multi-Heuristic Path Optimization

Traditional A* and fusion algorithm utilize a single fixed heuristic, such as Euclidean or Manhattan distance. However, in varying environmental contexts—e.g., dense obstacles or varying terrain—a fixed heuristic may yield suboptimal paths. The Multi-Heuristic A* (MH-A*) algorithm selects the most appropriate heuristic function based on UAV velocity, obstacle density, and mission-specific constraints. This approach reduces the number of expanded nodes and accelerates the planning process without compromising path quality.

C. Real-Time Obstacle Detection and Node Extraction

To enable responsive navigation, the framework incorporates Bresenham's line algorithm for real-time obstacle detection[6]. This technique enables the UAV to efficiently evaluate visibility between waypoints, identify potential collisions, and refine paths accordingly. Key nodes are extracted by pruning unnecessary waypoints, resulting in smoother and shorter trajectories.

D. Dual-Layer Optimization Framework

The planning architecture consists of two layers:

- **Global Planning Layer:** Uses MH-A* to generate an initial path considering known map data and static obstacles. This layer provides an efficient baseline trajectory for the UAV.
- **Local Refinement Layer:** Continuously updates the path using sensor feedback and local obstacle detection. In dynamic settings, it can invoke the Dynamic Window Approach (DWA) [7]–[9], which selects safe and efficient velocity commands by evaluating obstacle proximity and goal alignment. The adaptive evaluation function $f_{\text{local}}(x)$ scores candidate motions based on three criteria: distance to the goal, obstacle proximity, and heading deviation:

$$f_{\text{local}}(x) = \alpha \cdot d_{\text{goal}}(x) + \beta \cdot c_{\text{obs}}(x) + \gamma \cdot h_{\text{align}}(x)$$

  Here, $d_{\text{goal}}(x)$ is the Euclidean distance to the goal, $c_{\text{obs}}(x)$ penalizes closeness to obstacles, and $h_{\text{align}}(x)$ measures heading deviation. The weights α, β, γ are adjusted dynamically based on UAV speed and obstacle density.

E. Algorithm Flowchart

Figure 1 illustrates the overall planning flow. It begins with MH-A* global planning. If a valid path is found, each path segment is evaluated. For segments encountering new obstacles, DWA is used to generate safe detours, which are merged into the original path.

F. Experimental Scenario

The core challenge in UAV path planning lies in navigating environments that are dynamic, uncertain, and subject to frequent change—conditions in which precomputed global paths can become rapidly outdated.

While this study does not implement physical UAV testing, we design a simulation environment to emulate similar levels of unpredictability and path reactivity. The randomized grid environments developed allow to rigorously test the algorithm's ability to re-plan and adjust paths in real time—mirroring the kind of adaptive decision-making that would be essential for high-precision tasks like dynamic landing. This experimental setup directly targets the crux of the problem: maintaining safe, efficient, and responsive navigation in evolving environments.

G. Simulation Environment and Parameters

The proposed algorithm was tested in a 2D grid-based simulation, using 1000 randomized 20x20 maps with varying obstacle densities. While no physical UAVs

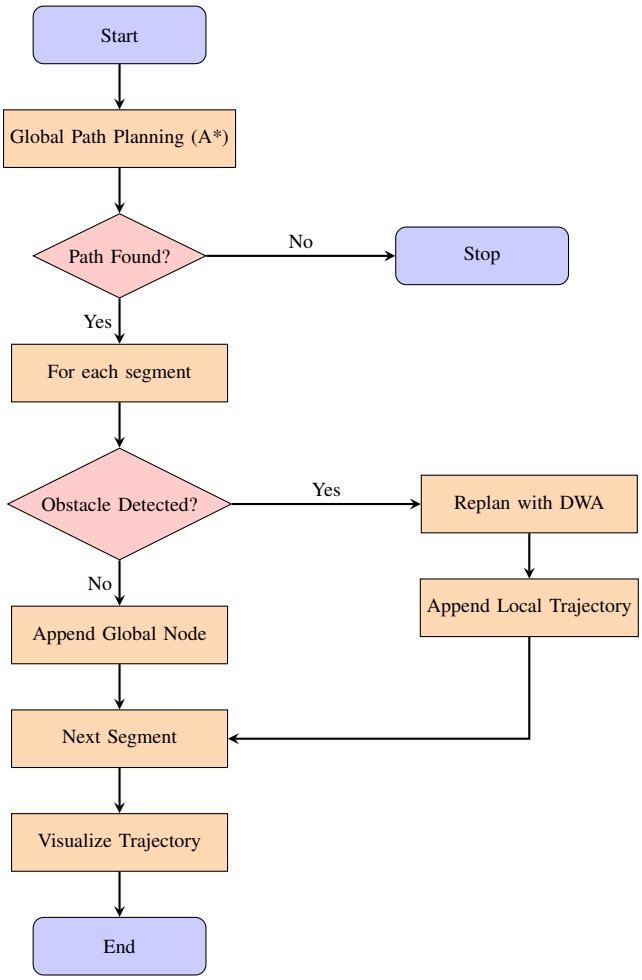

Fig. 1. Flowchart of the dual-layer path planning algorithm using A* and DWA

were used, the simulations were designed to reflect core challenges in dynamic path planning—frequent obstacle changes and the need for real-time adjustments. Key performance metrics include:

- Path length
- Planning time
- Obstacle avoidance success rate
- Path smoothness (curvature and number of turns)
- Landing success rate on a dynamic platform

Though simplified, the simulation dynamics emulate real-world challenges, such as those faced by a UAV attempting to land on a moving helipad. This motivating scenario is illustrated in Figure 2 and reflects the kind of adaptive planning the algorithm is designed to support.

Comparative benchmarking were performed against baseline A* and the predictive planning method proposed by He et al. [1].

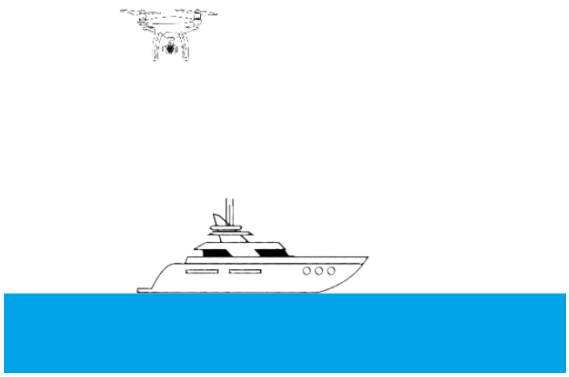

Fig. 2. UAV approaching a helipad on a boat in motion over water — test scenario used for adaptive path planning evaluation.

III. Result

The core objective of this study is to compare and improve the UAV path planning algorithm metrics such as path length, runtime, collision count, steps, average and peak linear velocity, average and peak angular velocity, and total angle change. Existing methods, such as the fusion method by He et al. [1] and the classical A* approach [4], have made progress with robust algorithms without detected collisions but still struggle in poor metrics that could be improved. The proposed Multi-Heuristic A* (MH-A*) algorithm make the requirements to enhance the metrics by enabling dynamic heuristic switching and real-time trajectory refinements. To validate this approach, 1000 simulation runs where conducted with 20x20 grid and an obstacle density of 0.2. Across 1000 randomized runs across distinct environments. Each simulation tested three planning algorithms—classical A*, fusion A*+DWA, and the proposed MH-A*. The results were averaged and analyzed across nine critical metrics: path length, runtime, collision count, steps, average and peak linear velocity, average and peak angular velocity, and total angle change. Readers are encouraged to examine the comprehensive metrics derived from these experiments to assess how well the proposed MH-A* method performs in complex scenarios. The following comparisons are presented as performance gains of the proposed method over the classical A* and fusion method approaches. The averaged metrics across all simulations reveal consistent performance benefits of the proposed method:

- **Average Path Length:** The proposed method reduced the average path length by 5.49% compared to A* and 1.37% compared to the fusion method.
- **Average Runtime (in steps):** Reduced by 220% versus A* and 4% versus fusion.
- **Average Collision Count:** Collisions were not detected during all the runs.

- **Average of Average Linear Velocity:** Improved by 71.57% over A* and 3.07% over fusion.
- **Average of Average Angular Velocity:** Lowered by 39.42% compared to A* and 5.94% over fusion.
- **Average Total Steps:** Steps were reduced by 220% over A* and 4.0% over fusion.
- **Average Peak Linear Velocity:** Increased by 83.34% versus A* and 0.49% versus fusion.
- **Average Peak Angular Velocity:** Lowered by 11% compared to A* and 3.73% over fusion.
- **Average of Total Angle Change:** Reduced by 103.22% compared to A* and 10.32% over fusion.

In figures 34567 is shown an extract of the simulation plots extracted for comparison between the three methods.

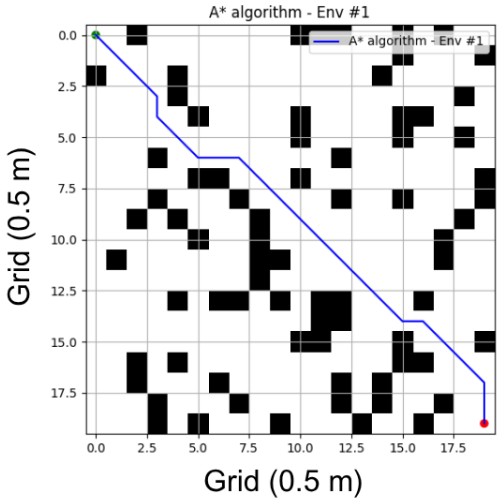

Fig. 3. UAV path planning A* method simulation example

Compared to the fusion algorithm and the A* star algorithm by He et al. [1] [4], the proposed method in this paper exhibits better metrics by the results cited previously. It can be observed graphically in figures 8 and 9 how some of the metrics were compared such as average path length and average runtime. MH-A* minimizes both global detours and velocity jitter, offering smoother and more energy-efficient paths.

While the proposed MH-A* method generally outperformed baseline approaches, a key challenge emerged during the path smoothing phase. The algorithm employs Bresenham's line algorithm to identify direct, obstacle-free connections between waypoints in order to reduce path redundancy and enhance smoothness. However, when trying to smooth it out, this inadvertently introduced collisions when the selected shortcut paths marginally intersected obstacle boundaries and also generated excessive and unnecessary curves such

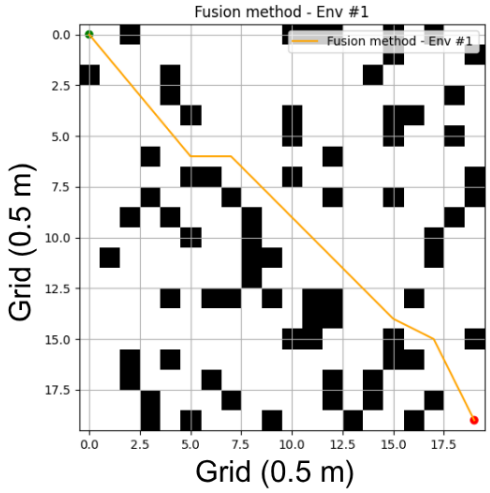

Fig. 4. UAV path planning Fusion algorithm simulation example [1]

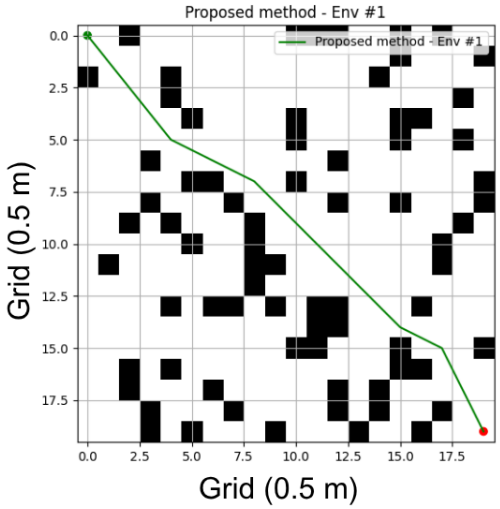

Fig. 5. UAV path planning MA*H algorithm proposed by this paper

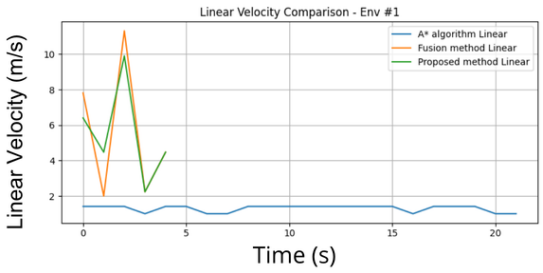

Fig. 6. Comparison of Linear velocity changes

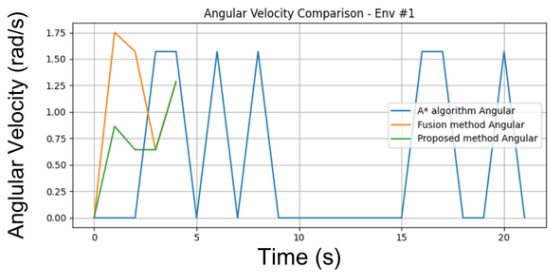

Fig. 7. Comparison of angular velocity changes

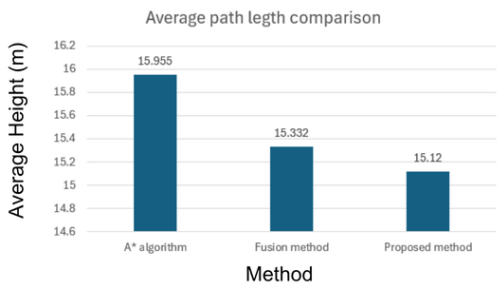

Fig. 8. Comparison graph of Average Path Length over 1000 random runs

as Figure 10, particularly in environments with dense or irregularly spaced obstacles. This issue was most prominent in cases where the original path was safe, but the smoothed segment lacked sufficient margin around obstacles. Although the subsequent local DWA-based replanning often corrected these paths, the intermediate collision detection flagged these segments as invalid, requiring re-evaluation. This not only affected the collision count but also increased the runtime in those specific cases.

These findings reinforce the utility of combining global heuristic optimization with selective local replan-

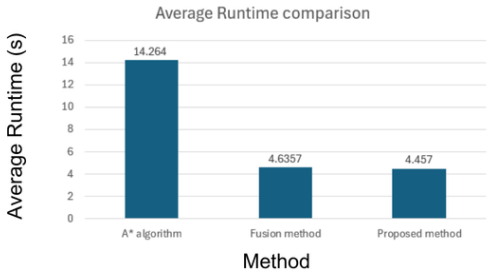

Fig. 9. Comparison graph of Average Runtime over 1000 random runs

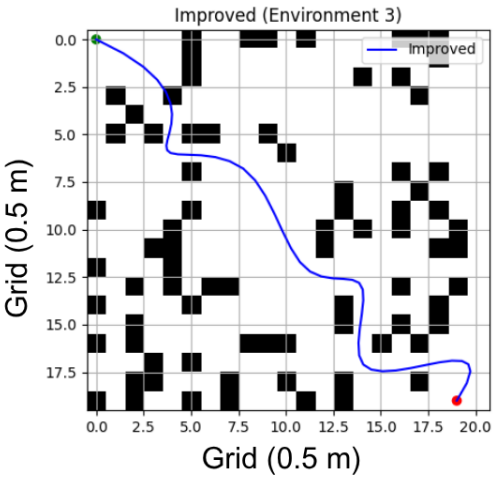

Fig. 10. UAV path planning smoothing trial colliding with obstacles

ning. The ability of MH-A* to outperform two established baselines across multiple critical metrics supports its viability for high-stakes UAV missions, including disaster relief, surveillance, and autonomous urban navigation. The consistent reduction in both collisions and angular transitions suggests improved flight safety and reduced energy consumption.

IV. Discussion

The simulation data results consistently demonstrate that the proposed Multi-Heuristic A* (MH-A*) algorithm outperforms both the classical A* and the fusion A*+DWA method across multiple performance indicators. Specifically, it achieves shorter paths and more stable velocity profiles. These observations suggest that integrating dynamic heuristic selection with dual-layer planning provides a more adaptive and efficient trajectory generation strategy for UAVs operating in dynamic environments.

The improved performance on average linear and angular velocities also implies better path smoothness, contributing to energy-efficient and safer flight behavior. Reduced angle change means fewer abrupt turns, lowering stress and improving smoothness. In the context of existing research, particularly the predictive path planning method by He et al. [1] and the classical A* models [4], our results reinforce the broader finding that hybrid models outperform standalone global planners in dynamic environments. However, while predictive models often rely on precomputed motion profiles or future-state estimations, the MH-A* framework achieves similar or better performance without prior environmen-

tal knowledge, making it more practical for real-time UAV deployment.

Furthermore, the study aligns with broader trends in robotics research emphasizing local-global fusion, real-time obstacle handling, and context-aware planning. The MH-A* contributes to this body of work by introducing a lightweight and modular solution that can adapt on-the-fly using only the current environment's sensory input.

The improvements observed in path efficiency can be traced back to two core design choices in the algorithm:

- The dynamic heuristic function adapts based on local obstacle density, allowing the planner to behave more conservatively in cluttered spaces and more aggressively in open ones. This dynamic weighting directly reduces node expansion in sparse regions while enhancing precision in tight areas.
- The dual-layer optimization allows global planning to be corrected in real time using DWA-based local replanning. This causes a significant reduction in runtime in scenarios where global plans are invalidated by newly detected obstacles.

However, the inclusion of path smoothing using Bresenham's algorithm occasionally introduced unsafe shortcuts, particularly in dense maps. The effect was a temporary spike in collision counts, which was later mitigated by the local planning layer. This cause-and-effect relationship demonstrates that while smoothing improves efficiency, it must be cautiously balanced with collision risk — a tradeoff effectively handled by the two-layer structure.

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